Course 50.050/50.550
Advanced Algorithms

Week 3 – Cohort Class L03.03

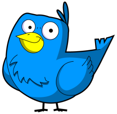
SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

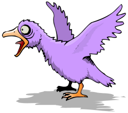
Friends and Strangers - Discussion

Question

Among the 6 of them, can we always find 3 of them that are either all friends with each other or all strangers with each other?

Friends and Strangers - Discussion





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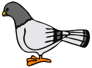

Yes! By the pigeonhole principle!

Friends and Strangers - Discussion

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Friends and Strangers - Discussion

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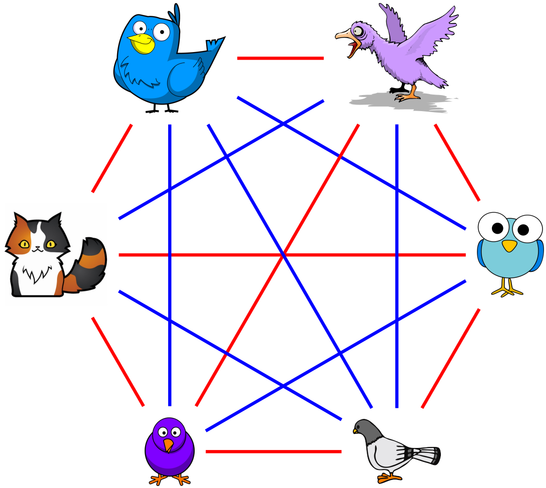
Friends

Strangers

Between every pair of “birds”, we can draw a **blue** edge for **friends**, or a **red** edge for **strangers**.

Friends and Strangers - Discussion



> **Question**
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Friends (blue line)
Strangers (red line)

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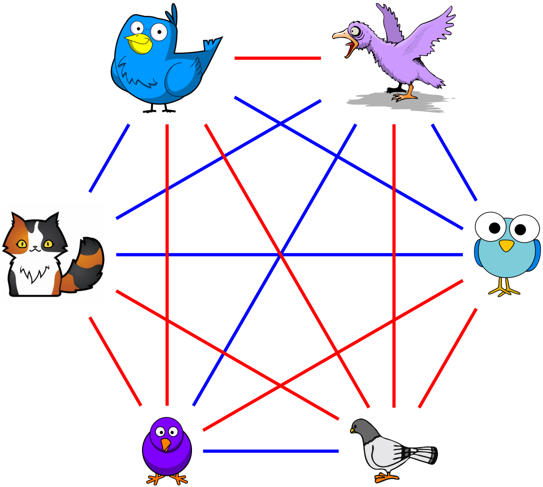
Friends and Strangers - Discussion



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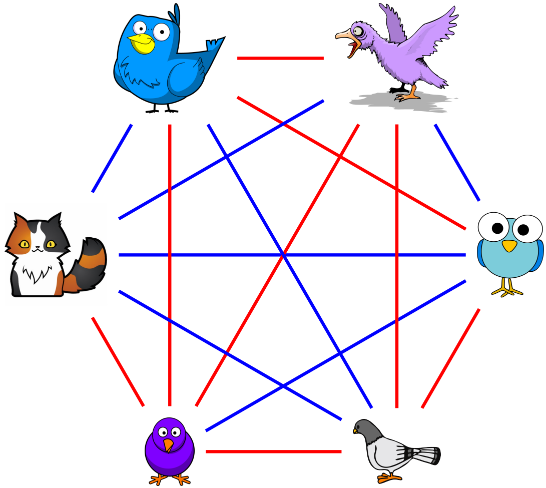

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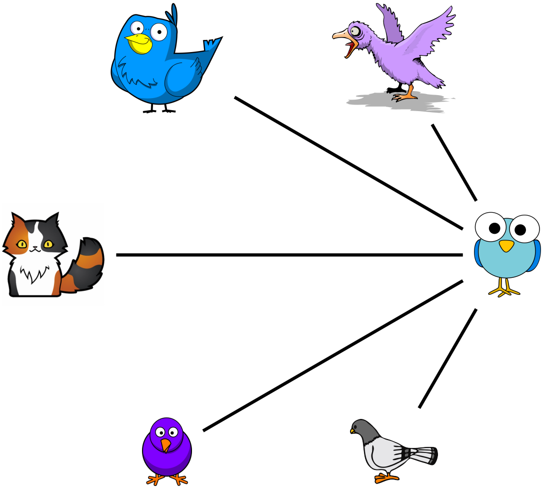

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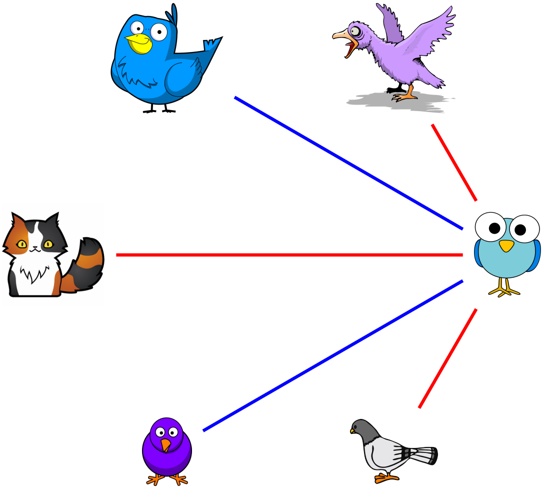
Friends

Strangers

Let's look at the edges incident to .

There are 5 other "birds" that  can be friends or strangers with.

Friends and Strangers - Discussion



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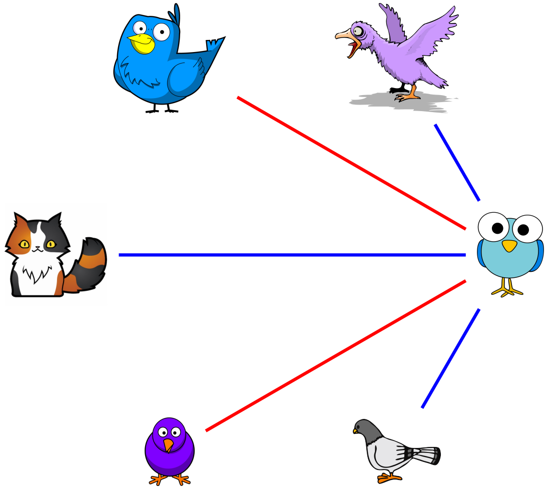
Among the 6 of them, can we always find 3 of them that are either all friends with each other or all strangers with each other?

Friends

Strangers

We have no idea who are friends or strangers with .

Friends and Strangers - Discussion



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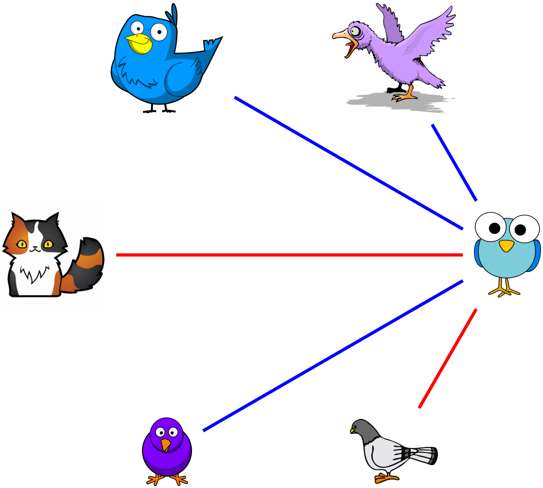

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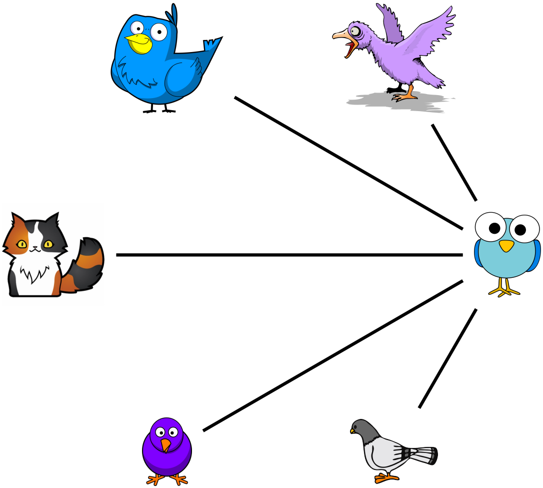

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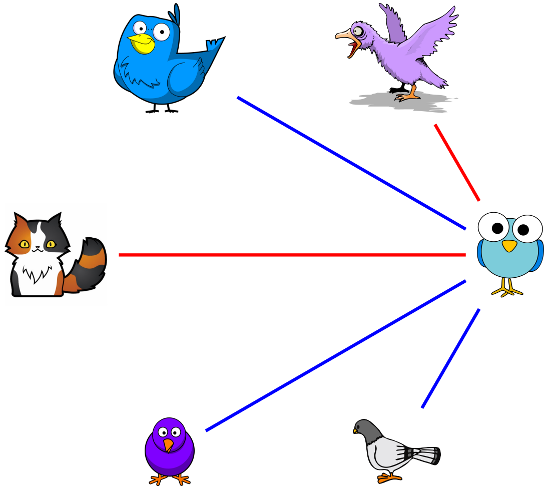
Friends

Strangers

But by the **pigeonhole principle**, at least 3 of these 5 edges have the same color.

Friends and Strangers - Discussion



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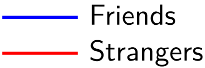
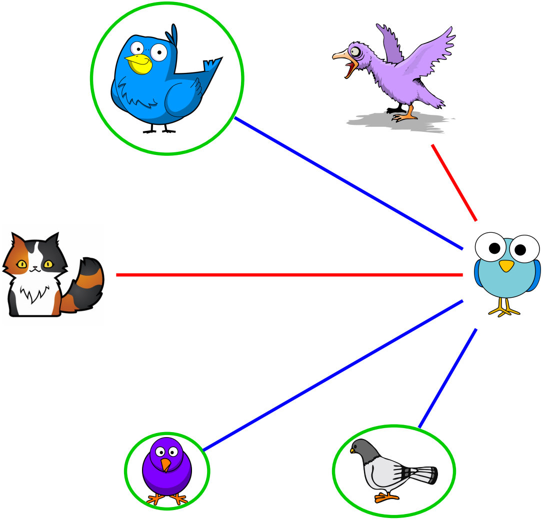


Suppose 3 of these 5 edges are colored blue.

Friends and Strangers - Discussion



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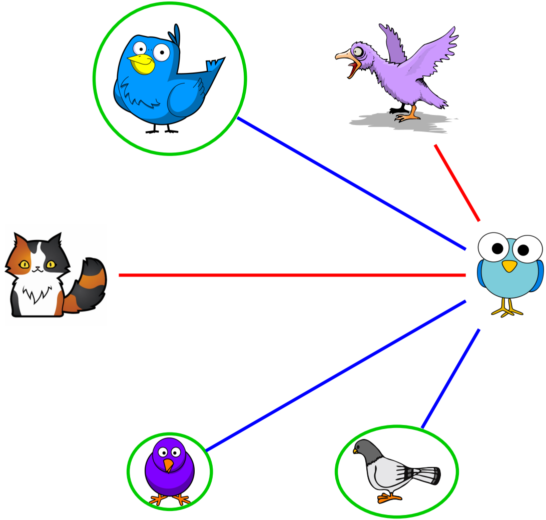
Friends

Strangers

Look at the 3 neighbors of  in this graph joined by blue edges.

Friends and Strangers - Discussion



Question

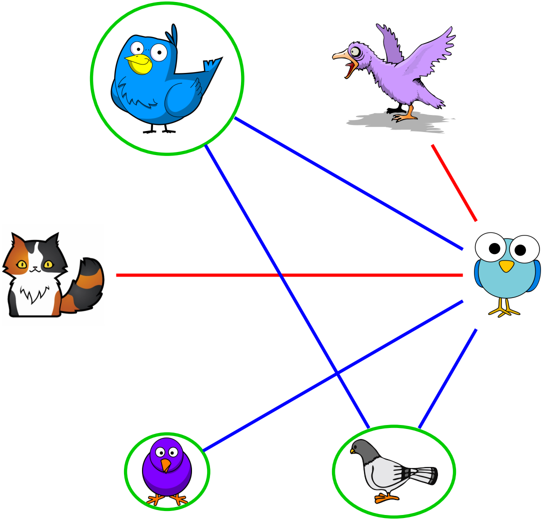
Among the 6 of them, can we always find 3 of them that are either all friends with each other or all strangers with each other?

Friends

Strangers

What happens when any two of them are friends?

Friends and Strangers - Discussion



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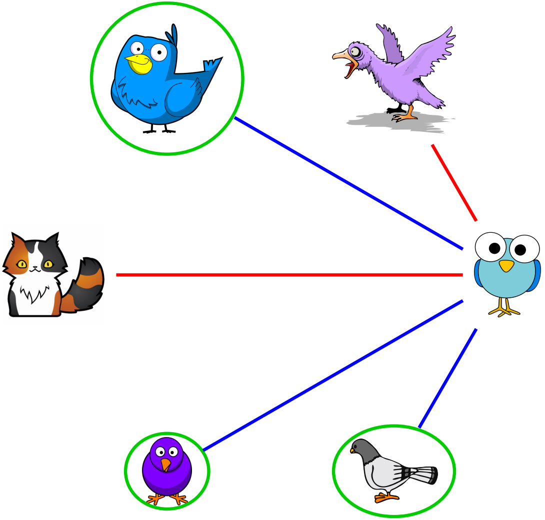
Friends

Strangers

For example, if  and  are friends, then    are all friends.

Friends and Strangers - Discussion



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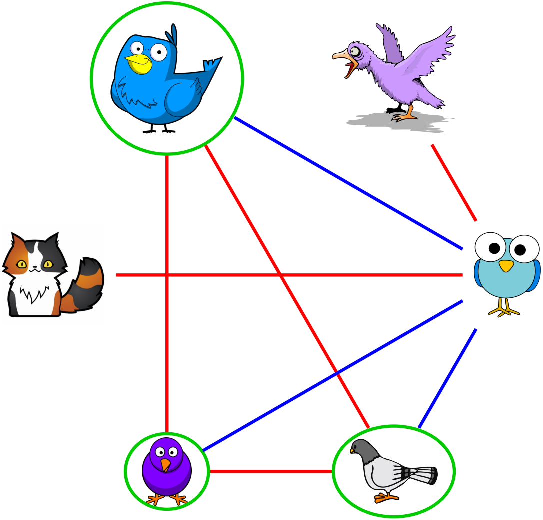
Friends

Strangers

So if any two of our three neighbors  are friends, then the answer to our question is yes!

Friends and Strangers - Discussion



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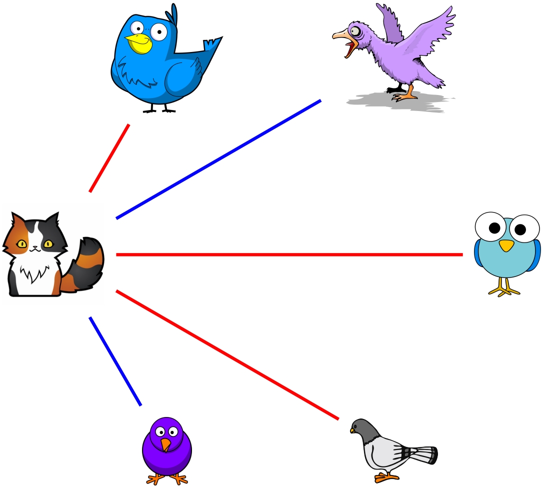
Friends

Strangers

Now, if no two among  are friends, then all 3 of them are strangers with each other! Again, our answer to the question is yes!

Friends and Strangers - Discussion



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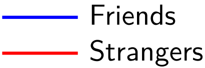
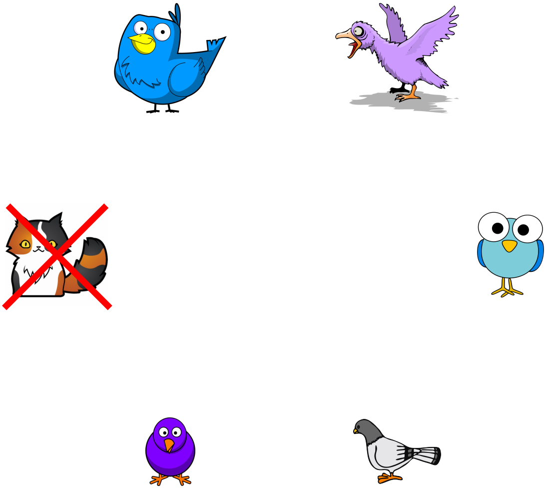


Remember that it doesn't matter which "bird" we begin with. Whichever "bird" we choose, the pigeonhole principle tells us that this "bird" will always have 3 incident edges of the same color.

Friends and Strangers - Discussion



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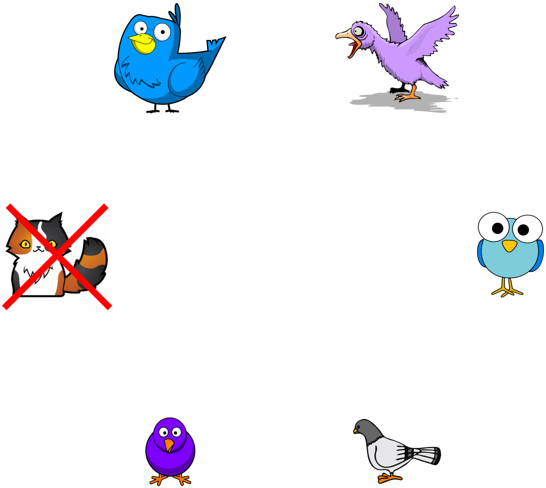
Friends

Strangers

What happens if we only have 5 (real) birds?

Friends and Strangers - Discussion



Question

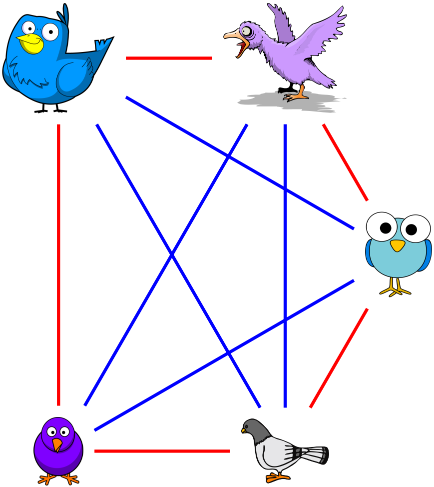
Among the **5** of them, can we always find 3 of them that are either all friends with each other or all strangers with each other?

Friends

Strangers

Can we always find 3 of these 5 birds that are either all friends with each other or all strangers with each other?

Friends and Strangers - Discussion



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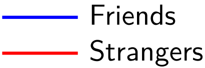
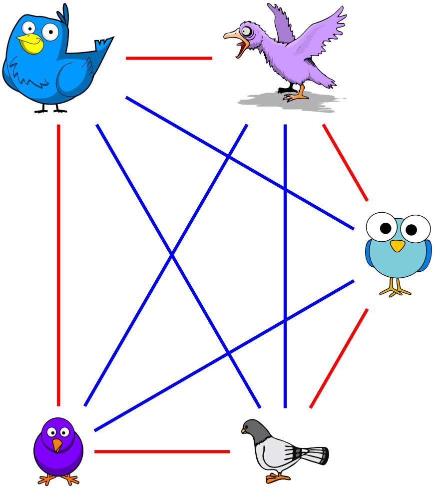


We can color the edges between these birds as shown here. There are no red triangles or blue triangles! So the answer is no!

Friends and Strangers - Discussion



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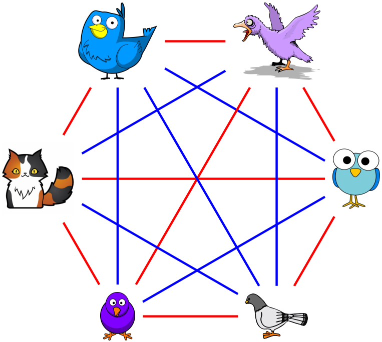
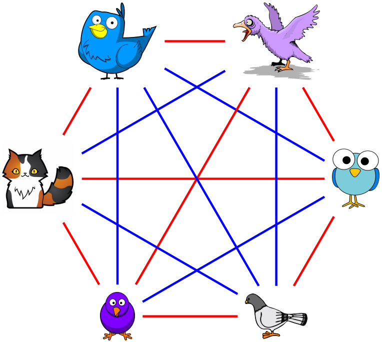


What we have shown

In any group of "birds", if the total number of "birds" is **at least 6**, then we can always find either 3 of them who are all friends with each other, or 3 of them who are all strangers with each other.

Friends and Strangers



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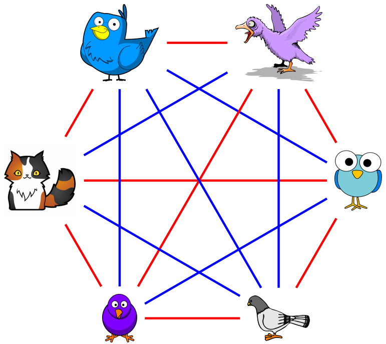
This **minimum** total number of "birds" for always finding **3** mutual friends or **3** mutual strangers is written as $R(\mathbf{3}, \mathbf{3})$.

Friends and Strangers



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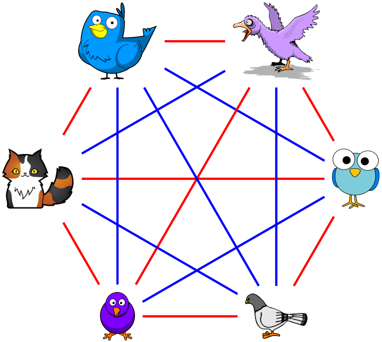
In other words, we have shown that $R(3, 3) = 6$.

Friends and Strangers



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Question: What about having 4 of them who are all friends with each other? Or 5 of them who are all strangers with each other?

Ramsey's theorem

Theorem: (**Ramsey's theorem**) Let $r, s \in \mathbb{Z}^+$. In a **sufficiently large** community of people, no matter who are friends or who are strangers, there will always be either r people who are all friends with each other, or s people who are all strangers with each other.

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- You may need to have a very very large number of people to guarantee that you can always find r mutual friends, or s mutual strangers.

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- You may need to have a very very large number of people to guarantee that you can always find r mutual friends, or s mutual strangers.
- The minimum total number of people in the community for this to be always true is called a **Ramsey number**, and it is written as $R(r, s)$.